Short Interest Trend Prediction with Large Language Models

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ABSTRACT

This paper studies the problem of short interest trend prediction using large language models. To do so, we provide a formal task definition and create a dataset for this task. We conduct extensive experiments with various types of large language models in different settings of in-context learning. Our results show that large language models have gained knowledge pertaining to short sale interest trends, and providing examples is beneficial.

CCS CONCEPTS

• **Computing methodologies** → **Information extraction**; • **Social and professional topics** → **Computing and business**.

KEYWORDS

Short Interest Trend, Natural Language Processing, Large Language Model

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1 INTRODUCTION

Information is the most vital asset in capital market investment and trading. Traders make decisions based on the information they gather, especially when it comes to short selling. Unlike traditional trading, which involves buying low and selling high, short selling entails borrowing and selling stocks with the anticipation that their prices will decrease before repurchasing and returning the borrowed stocks. Short sellers' information collection and distribution significantly impact the real economy, stock market efficiency, financial decisions, and auditing [5, 17]. It is revealed that public information is extremely important to short sellers because they have advanced skills in processing public information. Other than traditional quantitative financial reporting, [18] short sellers prefer news and initiate more trading on such days. Knowing how this public information. Understanding how short sellers process information from the news is both interesting and crucial for research.

News title	Label
Cobalt International Energy *Hits 52-Week High* of $20.68	increase
Polaris CEO *Bullish* About Sales to Military (PII, HOG)	increase
Gabelli & Co. *Downgrades* Lindsay Corporation to Hold	decrease
Credit Suisse *Maintains* Underperform on Patterson Companies, Inc., Raises PT to $45.00	decrease

Figure 1: An example of our dataset.

On a daily basis, the financial realm experiences a deluge of information, offering investors a wealth of data to navigate. From the venerable insights of the Wall Street Journal to the rapid-fire updates on social media from market experts, a multitude of sources inundate screens with analytical reports, algorithmic predictions, and real-time alerts. Recognizing the pivotal role of succinct communication, news organizations meticulously curate titles to encapsulate the essence of complex stories. These titles act as windows into the main points, enabling readers to glean quick and meaningful insights amid global finance's dynamic and swiftly evolving landscape without the need for in-depth exploration of entire articles.

In our dataset, words and phrases associated with a potential change in future stock prices include *Hit*, *Bullish*, *Downgrades*, and *Maintains* in the news titles. These terms are more likely to be linked to shifts in short interest. Figure 1 illustrates the information extracted from news titles for predicting short-interest trends. For example, *Hits 52-Week High* and *Bullish* explicitly indicate that the stock price will bump up in the future. Sophisticated short sellers perceive this information as a sign of market overreaction, leading to an overvaluation of the stocks. Consequently, they engage in short selling, causing a rise in short interest. This pattern allows us to observe an increase in short interest. Likewise, the *Downgrades* and *Maintains* directly indicate that the stock price will decrease in the future. Likewise, instead of betting on stock price will continue to go down, the sophisticated short sellers identify that as a good opportunity to cover the short position; therefore, short interest is likely to increase in such instances.

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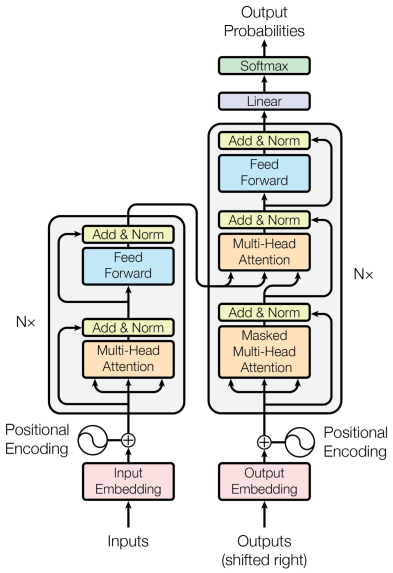


Figure 2: Architecture of the Transformer.

The research on short-interest trend prediction opens the door to many applications. One of them is in portfolio management, where accurate predictions of short interest trends can inform investment strategies and risk management. Financial institutions and hedge funds can leverage these predictions to optimize their portfolios, enhancing overall performance. Second, regulatory bodies may benefit from this research in monitoring and regulating short selling activities, contributing to the stability and integrity of financial markets. Third, traders and investors can use short interest trend predictions to make informed decisions, adjusting their positions based on anticipated market movements.

The main contributions of this paper are: 1) a novel task of short interest trend prediction, 2) a dataset for this task created by scraping the financial news related to public companies and their daily short sell data, 3) and experimental results with various LLMs in different settings of in-context learning showing that this challenging task can be automated by LLMs and providing examples before conducting inference is generally beneficial.

2 BACKGROUND

2.1 Short Interest Trend Prediction

Short-sellers are highly skilled and informed traders, and their activities affect stock market performance and efficiency[10]. Traditionally, finance researchers employ short interest, defined as *the number of shares shorted* over *total number of shares outstanding*, to measure the portion of shares to be shorted. The short interest serves as a predictor to predict future stock price underperformance ([11]). This phenomenon contributes to increased market efficiency and enhances the price discovery process for overvalued stocks. Detecting the change in short interest would help investors recognize the timely movement of short sellers and prepare for future investment and asset allocation.

Model input:

Review: Delicious food! Sentiment: Positive
Review: Terrible environment! Sentiment: Negative
Review: DIRTY PLACE!!! Sentiment: Negative
...
Review: Good meal and REALLY ENJOY THEM! Sentiment:

Model output:

Negative

Figure 3: An example of in-context learning.

2.2 Large Language Models

The objective of language model is to predict the next token based on the preceding tokens. In the pretraining phase, tokens following the current one are masked, and the model utilizes the prior tokens to predict the current token. Large language models (LLMs), are extremely scaled-up versions of original language models. They have at least 1 billion parameters and are typically pretrained on 0.1 trillion to 1 trillion tokens. LLMs can be categorized into two branches: one based on an encoder-decoder architecture and the other on a decoder-only architecture. In the context of LLMs, the encoder and decoder correspond to components in the Transformer architecture, as shown in 2. Each transformer block consists of multi-head self-attention layers, feed-forward networks, and skip connections. Representative works on encoder-decoder-based LLMs are T5 [9] and UL2 [14]. Representative works on decoder-based LLMs are LLaMA [15], LLaMA 2 [16], OPT [26], PaLM2 [1], and BLOOM [12]. Previous studies indicate that naive LLMs struggle to follow instructions and generate precise predictions. To address this, researchers have introduced instruction-tuning, a process where the parameters of naive LLMs are adjusted using carefully designed instructions for downstream tasks. FLAN is an influential work in instruction-tuning, featuring a collection of datasets consisting of 15M instances for 1,836 NLP tasks. After instruction-tuning on FLAN Collection [7], LLMs demonstrate their capability to comprehend instructions and produce accurate predictions.

2.3 In-Context Learning

In the context of LLM, in-context learning involves providing domain-specific or task-specific examples to LLMs before conducting inference [4]. Although it is not very clear, most researchers conclude that the improvement brought by in-context learning is from the ability to grasp the intricacies and nuances of particular subjects or industries. This adaptability ensures that the model remains relevant and accurate within specific contexts, as it can dynamically adjust its language generation patterns based on the input it receives. That said, if we provide a few examples before the question, LLMs can be "taught" to think about the problem more reasonably.

Figure 3 shows an example of in-context learning. To better elicit LLMs' ability to understand language, we can provide a few examples [1] before asking for the sentiment of the last review ("*Good meal and REALLY ENJOY THEM!*"). The presence of examples can "teach" LLMs to understand the review from the aspect of sentiment analysis.

[1]Note that the number of examples is typically determined empirically.

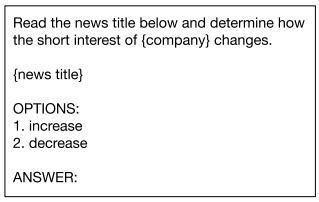


Figure 4: Our prompt. *company* is the name of the company we ask for the short interest trend.

3 DATASET CREATION

We choose to collect data from the Financial Industry Regulatory Authority (FINRA) [2] and Benzinga [3]. Specifically, we retrieve daily short sale data from FINRA, focusing solely on publicly traded companies listed on NASDAQ, NYSE, and AMEX. We work with these markets because they are US-based and significantly impact global trading markets. Our dataset spans from 2009 to 2020, excluding news published before 2009 due to scraping challenges and post-2020 trading data unavailability. For news titles, we employ an open web-scraping tool [4] to extract titles from Benzinga, excluding titles with over 50 tokens or any non-ASCII characters. Applying these filters could manage computational costs and align with most text processing APIs. Finally, our dataset comprises 136,192 news titles, each paired with the short interest trend of the associated company.

We create our dataset by matching the scraped news titles and daily short selling data. Specifically, we use the stock ticker of each publicly traded company and the timestamp of news as the primary keys to match the news title and the company's short selling data at the time when the news is published. We link the news title with the company's short selling data at the time of news publication, denoting the short interest on that day as S_t^c, where c and t represent the company and news timestamp, respectively. The short short interest trend of company c at time t is `increase` if $S_{t+1}^c - S_{t-1}^c > 0$. The short interest trend of company c at time t is `decrease` if $S_{t+1}^c - S_{t-1}^c < 0$. Note that we do not consider the case when $S_{t+1}^c - S_{t-1}^c = 0$ because it is very rare in our pilot study (less than 0.5% of all data).

4 DATASET ANALYSIS

54.7% of the 136,192 news titles in our dataset are labeled as `increase` and the remaining 45.3% labeled as `decrease`. We create stratified train and test splits (80% / 20%). The lengths of news titles in the training and test sets are similar (16.53 tokens vs. 16.48 tokens), but their vocabulary sizes differ significantly. The training data has 66,624 unique tokens, while the test data has 30,787 unique tokens. This difference in vocabulary usage allows the model to capture various aspects of financial knowledge.

5 EXPERIMENTS

We establish the majority label as the baseline. Among LLMs, we experiment with FLAN-T5 [2], FLAN-UL2 [13], and LLaMA 2 [16]. FLAN-T5 and FLAN-UL2 belong to the category of instruction-tuned encoder-decoder LLMs. Both models undergo instruction tuning on the FLAN Collection. As for LLaMA 2, we opt for LLaMA 2 Chat (7B), which is instruction-tuned on chat datasets specifically designed for dialogue-related tasks.

Figure 4 shows the prompt we use in our experiments with LLMs. We explicitly inform LLMs that we need them to determine how the short interest of the associated company changes. Following this, we provide the options from which we expect LLMs to select. Lastly, we use a flag ("ANSWER: ") to enable LLMs to make predictions by completing the prompts. We conduct experiments in different in-context settings by presenting 0/1/5/10 examples before providing our question. We do not experiment with more examples due to the maximum input lengths of these LLMs, ranging from 2k to 4k. Providing excessive examples would result in input lengths exceeding the models' limits, leading to truncation.

We use the 11B version of FLAN-T5 [5], which was released by Google. We use the 20B checkpoint of FLAN-UL2. [6] The LLaMA 2 Chat we use is released by Meta. [7] All LLMs are hosted by Huggingface [19]. To manage the computational cost, we set `truncation` as `True`, and the maximum input length is set as 2,048. [8] When loading the model, we use their 8-bit versions to control the GPU memory usage. Previous work shows that truncating the model parameter does not lead to performance degradation. [3] We conduct all experiments using Pytorch [8] and one NVIDIA GeForce RTX 3090.

6 RESULTS

Table 1 shows the weighted average F1 scores of LLMs in different settings of in-context learning. All LLMs outperform the majority baseline, indicating that this challenging task can be automated by LLMs. The best result (average F1: 0.47) is achieved by FLAN-UL2. The results reveal several insights.

- There exists a substantial disparity in performance between `increase` and `decrease`, suggesting that the financial knowledge acquired by LLMs during their pretraining process is imbalanced.
- Presenting more examples proves beneficial in most instances. For both FLAN-T5 and FLAN-UL2, the weighted average F1 scores show an improvement as the number of examples increases. Taking `increase` as an example, FLAN-T5's performance rises from 0.03 in the 0-shot setting to 0.31 in the 10-shot setting, and FLAN-UL2's performance increases from 0.18 in the 0-shot setting to 0.39 in the 10-shot setting. In contrast, LLaMA 2 does not gain from additional examples, as its F1 score drops from 0.17 in the 0-shot setting to 0.00 in the 10-shot setting. We attribute this to LLaMA 2's struggle with processing lengthy inputs effectively.
- Even the latest LLM, such as FLAN-UL2, struggles with this problem, achieving only an F1 score of 0.47. This contradicts the belief that LLMs can seamlessly handle any NLP task. To address this domain-specific challenge, we propose

[2]https://www.finra.org/finra-data/browse-catalog/short-sale-volume-data/daily-short-sale-volume-files
[3]https://www.benzinga.com/
[4]https://github.com/miguelaenlle/Scraping-Tools-Benzinga
[5]https://huggingface.co/google/flan-t5-xxl
[6]https://huggingface.co/google/flan-ul2
[7]https://huggingface.co/meta-llama/Llama-2-7b-chat-hf
[8]Recall that in the setting of 10-shot in-context learning, the model input could be very long.

	increase				decrease				Average			
	0-shot	1-shot	5-shot	10-shot	0-shot	1-shot	5-shot	10-shot	0-shot	1-shot	5-shot	10-shot
Majority	0.71				0.00				0.35			
FLAN-T5	0.03	0.23	0.30	0.31	0.62	0.59	0.60	0.59	0.30	0.40	0.38	0.44
LLaMA 2	0.17	0.06	0.00	0.00	0.40	0.17	0.14	0.31	0.28	0.08	0.10	0.39
FLAN-UL2	0.18	0.16	0.34	0.39	0.61	0.61	0.58	0.56	0.37	0.36	0.45	**0.47**

Table 1: Weighted average F1 scores of various LLMs in the different settings of in-context learning.

that LLMs should acquire more domain-specific knowledge during their pretraining phase. Furthermore, relying solely on pretraining LLMs for the task of next token prediction may be inadequate, as a comprehensive understanding of a company requires a broader foundation of commonsense knowledge in the finance domain.

7 CONCLUSIONS AND FUTURE WORKS

This paper introduces a new task—short interest trend prediction—and evaluates the performance of several large language models (LLMs) in this domain. To create the dataset, we collect financial news titles from Benzinga and daily short sell data from FINRA. We choose to work with FLAN-T5, FLAN-UL2, and LLaMA 2 since they represent both encoder-decoder and decoder-only architectures. We investigate the impact of in-context learning by varying the number of examples provided to each LLM. The findings reveal that LLMs can automate the task, providing more examples generally improves performance, and further refinements in pretraining are necessary to tackle our task.

In the future, we plan to investigate the role of instruction-finetuning in our task. Specifically, we plan to conduct experiments using Parameter-Effifficient Finetuning Methods (PEFT), such as LoRA [6], to further finetune the pretrained LLMs so that they can learn more domain-specific knowledge during the finetuning process. Additionally, we also plan to investigate whether and how the information carried by different modalities and other contextual content contributes to our task. Incorporating more than a single modality has been proven useful in many tasks, such as spatial information extraction [20], sentiment analysis [25], named entity recognition [23], and corporate event prediction [22]. Regarding the contextual content, it is also used in [21] and [24]. But to the best of our knowledge, taking into account multiple modalities and contextual content in short interest trend prediction still remains raw.

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